

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course : Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTETC503 - ANALOG COMMUNICATION

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following is not a basic component of a communication system?
A) Transmitter B) Receiver C) Loudspeaker D) Channel
- In simplex communication,
A) Data flows in both directions simultaneously B) Data flows in one direction only
C) Data flows alternately in both directions D) None of the above
- Which of the following is a pulse analog modulation technique?
A) PCM B) PWM C) TDM D) FDM
- 4 If modulation index is greater than 1, the AM wave is said to be:
A) Over-modulated B) Under-modulated
C) Critically modulated D) None of the above
- 5 Which AM type eliminates the carrier and retains both sidebands?
A) DSB-FC B) DSB-SC C) SSB D) VSB
- 6 Which AM variant is best suited for TV transmission?
A) DSB-SC B) SSB-SC C) ISB D) VSB
- Pre-emphasis is applied at the:
A) Receiver B) Transmitter C) Antenna D) Both A and B
- Which of the following is not a method of FM generation?
A) Armstrong method B) Reactance modulator C) Balanced modulator D) Varactor diode method
- Image frequency is defined as:
A) Signal frequency + carrier frequency B) IF + LO
C) LO + IF D) RF + IF
- 10 Which receiver stage suppresses variation in signal strength?
A) Mixer B) RF amplifier C) AGC D) IF filter
- 11 Thermal noise is directly proportional to:
A) Voltage B) Temperature C) Frequency D) Inductance
- 12 The unit of noise figure is:
A) Ohm B) Volt C) dB D) Hz

- Q2. Solve the following.
- A) Define Modulation. Explain necessity of Modulation in communication systems. 6
 - B) State and explain sampling theorem along with sampling process. A signal is sampled at 8 kHz. What is the maximum frequency that can be reconstructed? 6
- Q3. Solve the following.
- A) Compare DSB-FC, DSB-SC, and VSB modulation techniques. 6
 - B) Explain generation of SSB using phase shift method. 6
- Q4. Solve Any Two of the following.
- A) Define Frequency Modulation. Derive the mathematical expression for frequency modulated wave. 6
 - B) Explain the concept of pre-emphasis and de-emphasis. 6
 - C) Explain concept of phase modulation. For PM, if phase deviation is 2 radians and modulating signal is 5 kHz, find modulation index. 6
- Q5. Solve Any Two of the following.
- A) Explain essential characteristics of a radio receiver. 6
 - B) Explain the working of a superheterodyne receiver with the help of its block diagram. 6
 - C) What is double spotting and how is it avoided? 6
- Q6. Solve Any Two of the following.
- A) Define and explain thermal noise, shot noise, and transit-time noise. 6
 - B) Explain the working of a ratio detector in detail. 6
 - C) Describe the operation of a practical diode detector. 6

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